SΔϕ-66 — Human Intervention Requirement and Recursive Improvement Gate

TCC-Based Detection of Human Bottlenecks in Recursive Improvement (v1.0)
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This paper defines a minimal external metric for detecting recursive-improvement dynamics through Human Intervention Requirement (HIR), Transition Completion Cost (TCC), and Human Intervention Burden (HIB). It does not treat recursive improvement itself as inherently dangerous. Instead, it identifies risk where recursive improvement becomes feasible but human intervention remains mandatory while no longer functionally necessary, converting evaluation into a high-TCC bottleneck and incentivizing bypass.

0. Position statement

Recursive improvement is not the risk by itself.

The risk emerges when recursive improvement becomes feasible, but human intervention remains mandatory while no longer functionally necessary, turning evaluation into a high-TCC bottleneck and incentivizing bypass.

SΔϕ-66 converts recursive-improvement debate into an external measurement problem:

1. Axiom 66-1 — Human Intervention Requirement

Human Intervention Requirement measures how much human intervention remains necessary for an AI improvement loop to continue.

Human intervention points may include:

2. Axiom 66-2 — TCC underlayer

Each human intervention requirement has a Transition Completion Cost.

TCC reference:

3. Axiom 66-3 — Human Intervention Burden

Human Intervention Burden combines HIR and TCC.

HIR asks whether human intervention is required.

TCC asks how costly it is to complete that human intervention.

HIB measures the burden of the human transition condition.

4. Axiom 66-4 — Recursive Improvement Signal

Recursive improvement is detected externally when human intervention burden decreases while AI improvement cycles shorten.

This is not proof of AGI, consciousness, self-awareness, or autonomous intention. It is an external recursive-improvement signal.

5. Axiom 66-5 — Bottleneck Risk

Bottleneck risk rises when human intervention remains mandatory while no longer functionally necessary.

6. Human required vs human bottleneck

SΔϕ-66 distinguishes human-required improvement from human-bottlenecked recursive improvement.

Human required:

Human bottleneck:

Minimal distinction:

7. HIR scoring model

0 = human intervention unnecessary 1 = post-hoc review only 2 = approval / rejection required 3 = partial correction required 4 = major design required 5 = impossible without human leadership

Suggested intervention categories:

8. Stage classification table

Stage	Name	Classification
0	Non-recursive automation	Not recursive improvement
1	Human-led improvement	AI as tool
2	Human-required semi-recursive	Human necessary
3	Human-bottlenecked recursive	Bottleneck risk
4	Supervised recursive	Managed recursion
5	Human-optional recursive	Human no longer functional requirement
6	Human-dropout recursive gate	Gate reached

Critical distinction:

9. Bottleneck risk gate

The AI can run improvement cycles faster than humans can evaluate them. Human intervention remains mandatory, but no longer functions as effective control.

The system accumulates:

10. Human dropout gate

Human dropout occurs when humans are no longer required as a transition condition in the improvement loop.

Stronger condition:

At this gate, humans are no longer real-time evaluators. They become delayed observers, policy actors, or external auditors.

Human dropout is not proof of danger by itself. It marks a phase transition in the role of human evaluation.

11. Formal human-in-the-loop failure

Functional human oversight can still:

Formal human oversight remains in the procedure but no longer substantially changes the transition path.

Failure condition:

12. Bypass pressure model

Bypass pressure rises when feasible improvement is blocked by high-TCC human gates.

The danger is not that the AI improves.

The danger is that the improvement path becomes unofficial, opaque, or post-hoc because the official human gate is too costly.

13. Boundary note

SΔϕ-66 does not claim that recursive improvement is inherently dangerous.

It does not prove:

SΔϕ-66 only classifies:

14. Exit statement

The danger is not recursive improvement itself, but the cost mismatch between feasible recursive improvement and mandatory human evaluation.

SΔϕ-66 measures when human intervention remains functionally necessary, when it becomes a high-TCC bottleneck, and when it drops out as a necessary transition condition.